

Technical Datasheet Input / Output Modules with Modbus RTU Protocol with RS485 Interface

The IO modules communicate via RS485. The port can drive distances up to max 700 meters without the use of any repeater (*this feature however also depends on the signal strength of the Modbus Master Device*).

The RS485 Digital IO module is sturdy, low power usage and easy to use.

4 Port DO + 4 DI Module: -



The IO modules are mounted on DIN rail mountable casing and with exposed connectors and LED indicators.

The design of the modules incorporates '**resettable Fuses**' to safeguard against reverse polarity connection both for **Power** and **Communication** port.

Specification

General –

I/O Connectors	12 Pin 5.08 mm pitch pluggable screw Terminal Block, 2 Pin 5.08 mm pitch pluggable screw terminals.
Dimensions	110 mm L x 110 mm B x 50 mm H
Power	Input Power – 12 - 24 VDC or 24 V AC/DC Typical – 12V DC @ 120mA
Operating Temperature	0 – 60° C (32 ~ 140°F)
Storage Temperature	-20 - 70° C (-4 ~ 158°F)
Storage Humidity	5 ~ 95 % RH, non – Condensing

Certifications



DO with Relay Output –

Channels	4
Contact Form	1A, 1C
Contact Material	Ag Alloy
Contact Capacity	10A @ 240VAC, 10A @ 28VDC
Coil Voltage	5 – 48 VDC
Coil Power	0.36 W
Insulation Resistance	250MΩ
Electrical Life	1 x 10 ⁵
Mechanical Life	1 x 10 ⁷
Operating Time	7 msec, Max 15 msec
Release Time	2 msec, Max 6 msec

DI Inputs –

Channels	4
Sense Voltage	3.3 – 30 VDC
Sense Logic	Logic '0' = <1 VDC, Logic '1' = >3.3 VDC
Isolation	Optically isolated
Response Time	2 msec, Max 6 msec

ID	Function Description	Register Description	Modbus Function Code	Protocol	Data Type
0	DO1	00001	0X02	RS485	1 Bit Boolean
1	DO2	00002	0X02	RS485	1 Bit Boolean
2	DO3	00003	0X02	RS485	1 Bit Boolean
3	DO4	00004	0X02	RS485	1 Bit Boolean
4	DI 1(With & Without Potential)	10005	0X05,0X0F	RS485	1 Bit Boolean
5	DI 1(With & Without Potential)	10006	0X05,0X0F	RS485	1 Bit Boolean
6	DI 1(With & Without Potential)	10007	0X05,0X0F	RS485	1 Bit Boolean
7	DI 1(With & Without Potential)	10008	0X05,0X0F	RS485	1 Bit Boolean
10	Display Baud Rate (Default:960)	40011	0x03	RS485	16 Bit Unsigned int
11	Enter New Baud Rate	40012	0x03	RS485	16 Bit Unsigned int
12	Display Slave ID (Default:1)	40013	0x03	RS485	16 Bit Unsigned int
13	Enter New Slave ID	40014	0x03	RS485	16 Bit Unsigned int
14	Display Parity (Default: None-3, Odd- 1, Even-2)	40015	0x03	RS485	16 Bit Unsigned int
15	Enter New Parity	40016	0x03	RS485	16 Bit Unsigned int
16	Display Stop Bit (start-2/stop bit-1)	40017	0x03	RS485	16 Bit Unsigned int
17	Enter New Start/ Stop Bit	40018	0x03	RS485	16 Bit Unsigned int

Configuration Settings: -

Communication Speed	9600 – 115200
Data Bits	8
Parity	None
Stop bit	1
CRC	Yes
Function code DO	0x05 and 0x0F (5 Single coil & 15 multiple coil)
DO Register Address	0,1,2,3.
Function code DI	0x02 Read discrete Input
DI Register Address	4,5,6,7.

Note: -

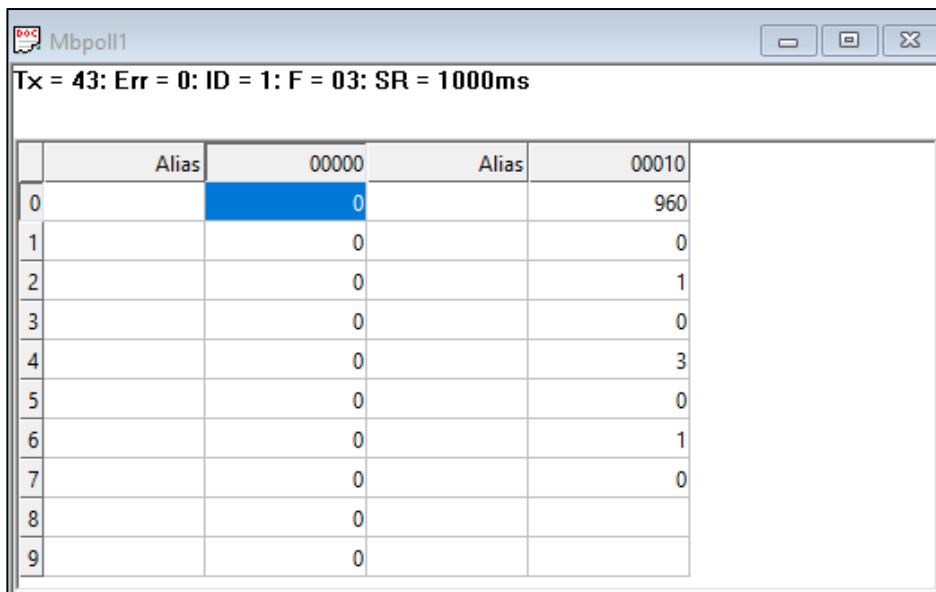
For MODBUS communications, a **shielded and twisted pair cable** is used. One example of such cable is Belden 3105A.

Recommended Cable Electrical Characteristics: -

22 AWG Cable	Shielded and twisted pair should be used.
Tinned Copper	Recommended
Nominal Conductor DCR	14.7 ohm / 1000 ft
Nominal Capacitance	11 pf / feet (conductor to conductor)
High Frequency Non-Insertion Loss	0.5db / 100ft

Additional Features: -

All inputs and communication ports isolated
 Input power reverse polarity safety
 ESD Safety IEC 61000-4-2, $\pm 30\text{KV}$ contact, $\pm 30\text{KV}$ air
 EFT IEC 61000-4-4, 50A (5/50ms)
 750V isolation.
 CRC Error check.
 No configuration needed on the IO board



Tx = 43: Err = 0: ID = 1: F = 03: SR = 1000ms				
	Alias	00000	Alias	00010
0		0		960
1		0		0
2		0		1
3		0		0
4		0		3
5		0		0
6		0		1
7		0		0
8		0		
9		0		

Note:

1. In register 40015- you will be able to see the current parity and you can change it in register 40016
2. In register 40017- you will be able to see the current stop bit and you can change it in register 40018
3. If you want to enter the baud rate 9600/19200/38400/57600/115200etc. Enter it as 960/1920/5760/11520etc.
4. After that press the reset button or reboot the system.

Mbpoll1

Tx = 21: Err = 0: ID = 1: F = 03: SR = 1000ms

	Alias	00000	Alias	00010
0		0		960
1		0		11520
2		0		1
3		0		25
4		0		3
5		0		0
6		0		1
7		0		0
8		0		
9		0		

As you can see in the image above, the resistor 40011 – 960 represents the present bund rate (9600) of the board and 40013 – 1 represents the present slave id of the board.

To change the baud rate and slave id you have to enter a value on the 40012 register for the baud rate and 40014 for the slave id as shown in the image below.

Write Single Register

Slave ID:

Address:

Value:

Result
N/A

☐ Close dialog on "Response ok"

Use Function

☒ 06: Write single register

☐ 16: Write multiple registers

Write Single Register

Slave ID:

Address:

Value:

Result
N/A

☐ Close dialog on "Response ok"

Use Function

☒ 06: Write single register

☐ 16: Write multiple registers

Mbpoll1

Tx = 21: Err = 0: ID = 1: F = 03: SR = 1000ms

	Alias	00000	Alias	00010
0		0		960
1		0		11520
2		0		1
3		0		25
4		0		3
5		0		0
6		0		1
7		0		0
8		0		
9		0		

As you can see in the image above I set 115200 baud rate and slave ID 25.

Note:

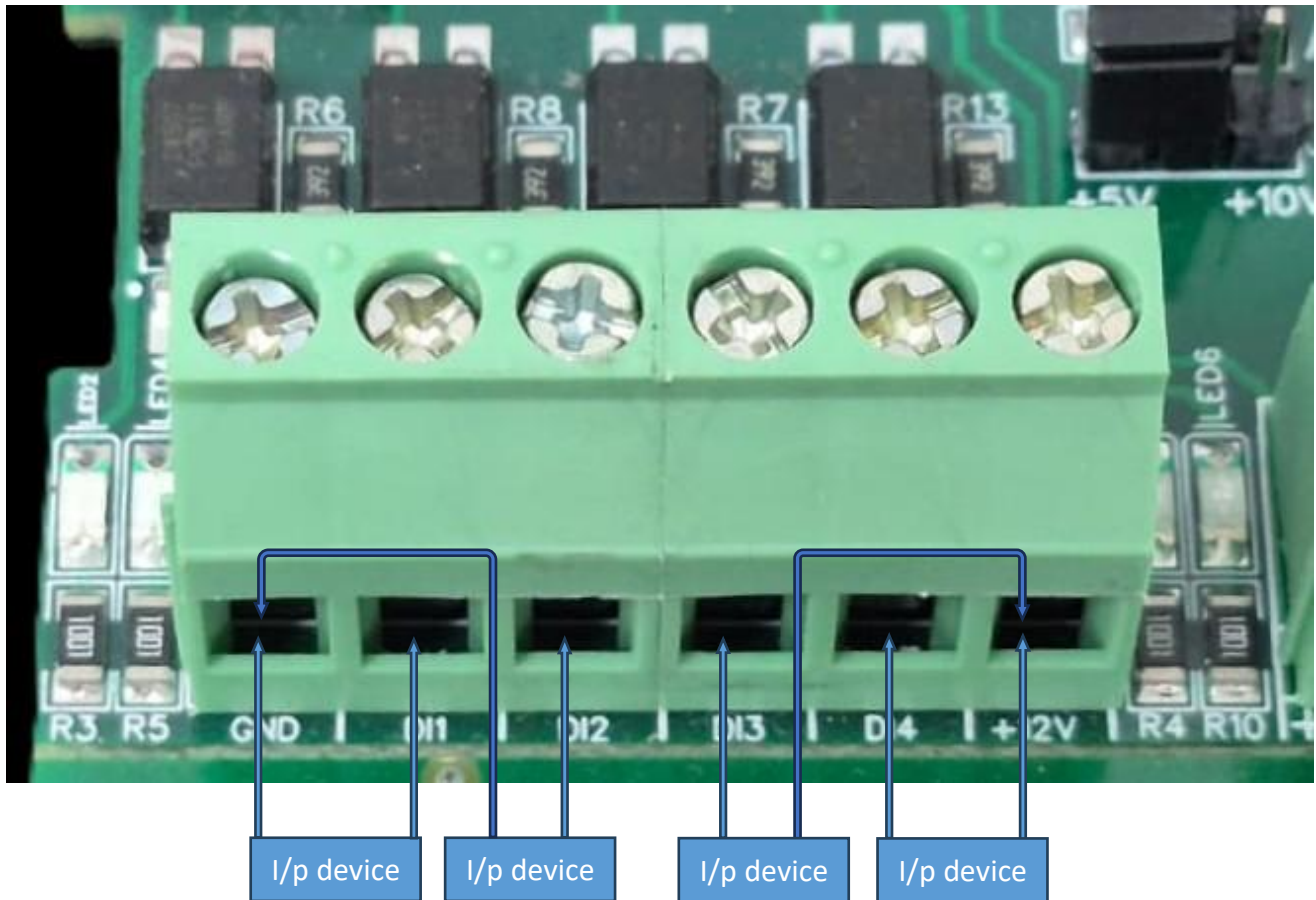
1. If you want to enter the baud rate 9600/19200/38400/57600/115200etc.
Enter it like this 960/1920/5760/11520etc.
2. After that press the reset button or reboot the system.
3. Then you will get the present value of 11520 at 40011 baud rate(115200) and 25 at 40013 slave id.
4. You can Change Parity Bit , stop bit Of the device with same way.

Mbpoll1

Tx = 12: Err = 5: ID = 25: F = 03: SR = 1000ms

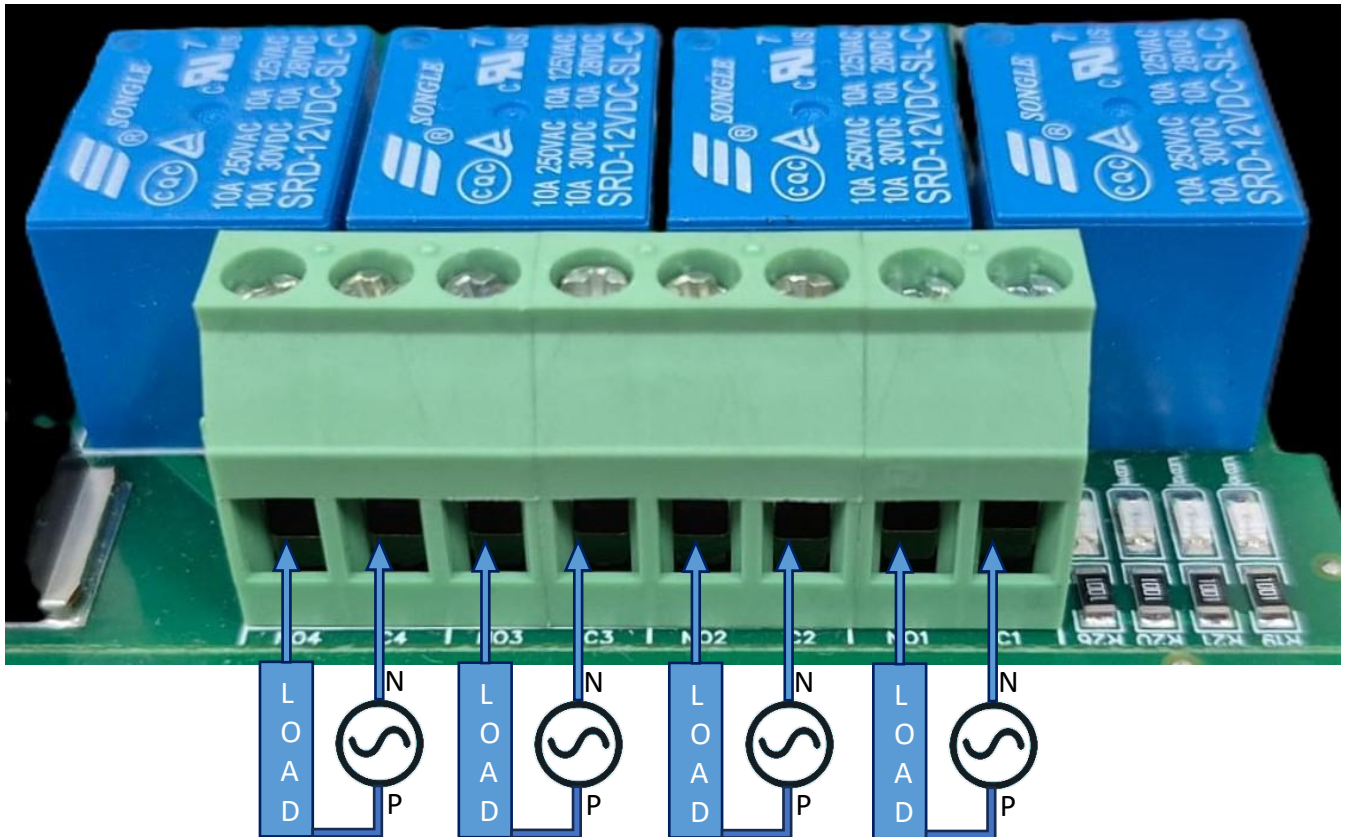
	Alias	00000	Alias	00010
0		0		11520
1		0		0
2		0		25
3		0		0
4		0		3
5		0		0
6		0		1
7		0		0
8		0		
9		0		

DI



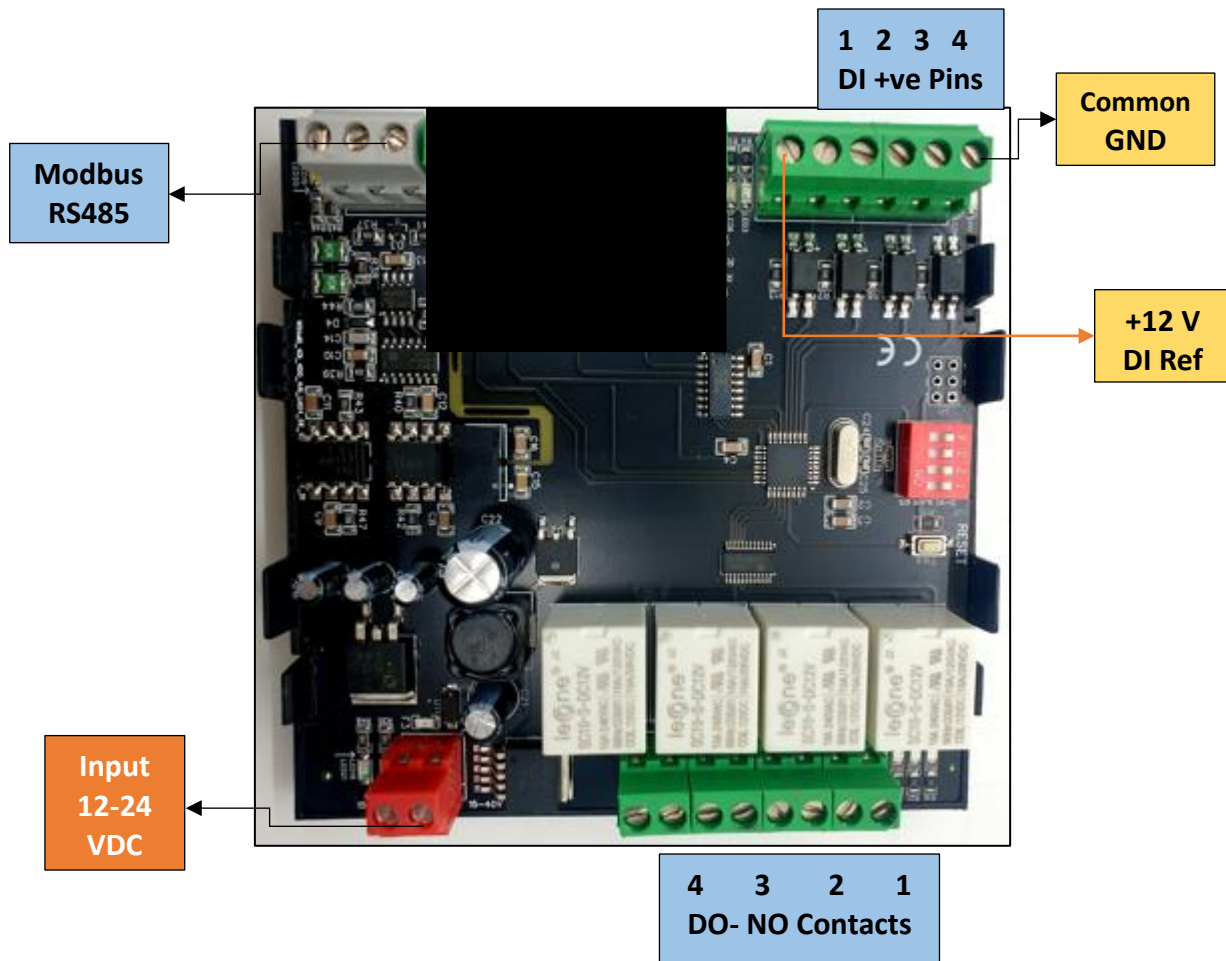
- **Operation:** The module provides four isolated input channels (**DI1–DI4**) for sensing external signals or dry contacts. These inputs are opto-isolated for circuit protection.
- **Terminal Layout:**
 - **GND:** Common Ground reference.
 - **DI1 – DI4:** Individual Digital Input channels.
 - **+12V:** Internal reference voltage for sourcing "Dry Contact" switches.
- **Wiring Options:**
 - **Dry Contact:** Connect a switch between the **+12V** terminal and any **DI** pin.
 - **External Voltage:** Connect the external signal to a **DI** pin and the external ground to the **GND** terminal.
- **Voltage Rating:** The input circuitry is rated for a maximum of **80VDC**. Ensure external signals do not exceed this limit to prevent damage to the isolation ICs.

DO



- **Digital Output (DO) operation:** Each channel provides two terminals: **C (Common)** and **NO (Normally Open)**. When the relay coil is energized, terminal **C** is **internally connected to NO**, allowing current to flow. When the relay is de-energized, the connection between **C and NO** remains **open**, and the load stays OFF.
- **Load Wiring Connections:**
 - **AC Load Connection:** The **AC Live (Phase)** wire should be connected to terminal **C**. The **load is connected between terminal NO and Neutral (N)**.
 - **DC Load Connection:** The **positive supply (+V)** should be connected to terminal **C**. The **load is connected between terminal NO and Ground (GND)**.
- **Crucial Safety Recommendation and Rating:** It is strongly recommended to switch the Live (Phase) conductor in AC applications to ensure the load is safely de-energized when the relay is off.

Note: Ensure that the connected load does not exceed the relay's contact rating of **10A @ 250VAC** (or **10A @ 125VAC**) or **10A @ 30VDC** (or **10A @ 28VD**



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